



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

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2026-10-02

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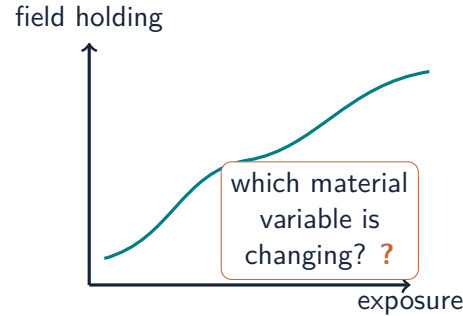
Sequel to last year's MeVArc talk. 2025 showed the dislocation model and cross-sectional STEM finding a modified near-surface layer; I closed asking for more comparative forensics—other materials, more samples. This year answers that: a large-area copper measurement testing the model's central prediction, and a stainless-steel experiment testing whether it generalizes. E_S is kept out of the introduction; appears once near the end.

Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.

Something in the metal stores the improvement. We do not know what.



Source: Wuensch et al., Rev. Mod. Phys. 98, 025004 (2026); Degiovanni et al., PRAB (2016). See also W. Wuensch, this morning (09:15).

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└ Conditioning works. What does it change?

Two macroscopic clues constrain the answer: the near-thirtieth-power field dependence, and conditioning following pulse count not breakdown count. Both say the applied field, not the arc, drives the work. Look for a field-driven change in the metal itself.

Conditioning works. What does it change?

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field holding

exposure

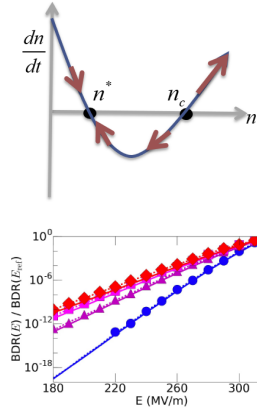
which material variable is changing? ?

Source: Wuensch et al., Rev. Mod. Phys. 98, 025004 (2026); Degiovanni et al., PRAB (2016). See also W. Wuensch, this morning (09:15).

Our model: conditioning as collective dislocation dynamics



- ▶ Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- ▶ One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



Central prediction: conditioning *evolves the near-surface dislocation structure*.

Source: Engelberg et al., PRL 120, 124801 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020).

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└ Our model: conditioning as collective dislocation dynamics

MDDF model is our own (2018–2020). Cartoon shows birth-death dynamics of the mobile-dislocation population: below critical density it decays, above it a runaway triggers nucleation. The plot reproduces the field dependence of breakdown rate. Key line for this talk: conditioning should leave an evolving dislocation structure just below the surface. Everything after this slide tests that sentence.

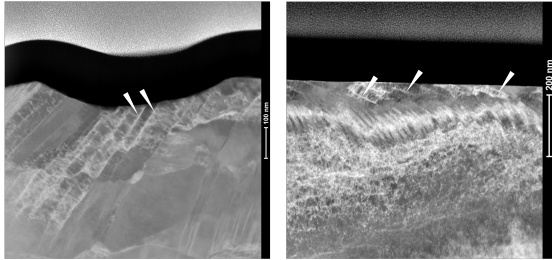
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STEM already saw it, at single spots



Cross-sectional STEM of conditioned hard (as-machined)
Cu: dislocation-poor zone in the top ~ 200 nm.

Seen at one spot. The large-area comparison was still missing.

- ▶ **Direct** evidence of near-surface plasticity in conditioned regions.
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

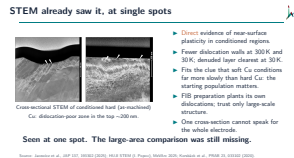
Source: Jacewicz et al., JAP 137, 193302 (2025); HUJI STEM (I. Popov), MeVArc 2025; Korsbäck et al., PRAB 23, 033102 (2020).

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Does conditioning leave a structural memory?

└─ STEM already saw it, at single spots

2025 STEM images. Dark band is the protective coating; just beneath it the field-exposed region is nearly free of dislocation walls while the reference keeps walls continuous to the surface. Published wording is conservative: 300 K shows significant reduction of walls; the denuded zone is clearly visible at 30 K. The paper gives no thickness; 200 nanometers is read from images, the number the manuscript uses. Hard-versus-soft copper contrast (Korsbäck 2020) is an independent macroscopic clue: hard copper conditioned within about 25 million pulses; soft samples began saturating near 600 million or stayed unsaturated longer, consistent with different starting dislocation populations. Two lessons: FIB preparation generates dislocations and loops, so only larger-scale structure is trustworthy; TEM and STEM is intrinsically local. Last year I asked for more comparative forensics: other materials, more samples. That is exactly the gap next slides fill.



A tiny stress, repeated a billion times

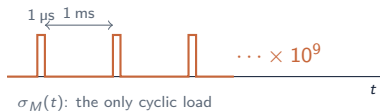


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa**

$\sim 10^3 \times$ below the yield of annealed Cu



The model predicts a large-area signature. Last year we asked for it; this year we looked.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007). Field-driven deformation at the nanoscale:

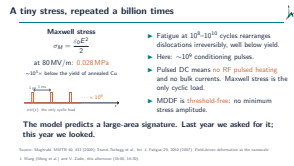
J. Wang (Meng et al.) and V. Zadin, this afternoon (16:00, 16:30).

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Does conditioning leave a structural memory?

└ A tiny stress, repeated a billion times

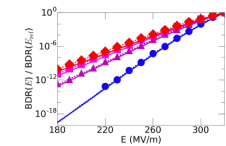
Maxwell stress is three orders of magnitude below yield. The point is not amplitude but accumulation. VHCF shows oscillatory loading well below yield still rearranges dislocations irreversibly for a billion cycles—our regime. Pulsed-DC is cleaner than RF structures: no RF magnetic field, no pulsed surface heating, no bulk Ohmic current. Maxwell stress is the only cyclic mechanical load. Earlier RF-based studies could never fully separate thermal fatigue from field effects. MDDF is threshold-free: does not need the stress to exceed any yield-like value. That argument bridges the 2025 model-and-microscopy story to this year's large-area measurement.



Last year we asked for more forensics. Here it is.

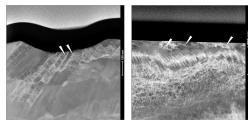


MeVArc 2025



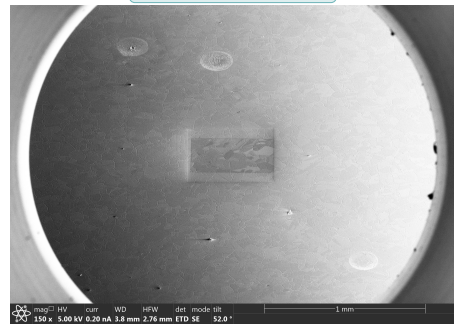
MDDF model

Closed with: other materials? more samples?



local STEM denuded zone

MeVArc 2026



Large-area EBSD across one cathode

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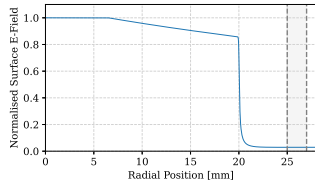
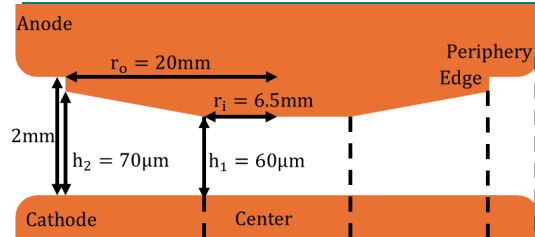
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└ Last year we asked for more forensics. Here it is.



Light visual bridge: 2025 showed the model and local cross-section; 2026 delivers the large-area structural comparison on one electrode. Do not quote the 75 percent effect here.

One cathode turns field exposure into a controlled coordinate



- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1 μ s pulses at 1 kHz
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

The field gradient gives us internal controls on a single specimen.

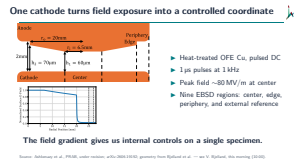
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; geometry from Bjelland et al. — see V. Bjelland, this morning (10:00).

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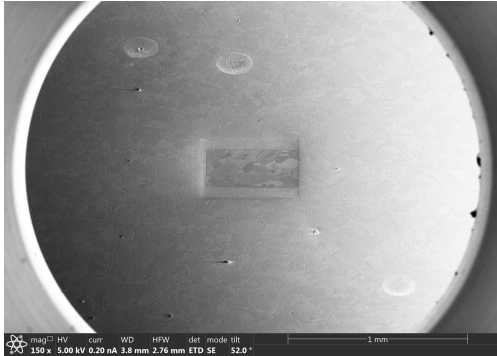
Does conditioning leave a structural memory?

└ One cathode turns field exposure into a controlled coordinate

The sloped anode maps radius onto local field: highest at flat center, decaying across the sloped gap, below about 2.5 MV/m beyond the anode radius. Comparing positions on one cathode removes cathode-to-cathode preparation differences. Separate identically treated but unexposed cathode gives an external reference. That geometry is the "more comparative forensics" design from last year's outlook.



EBSD maps plastic activity far from breakdown craters



Cleaned ROI is hundreds of microns from visible craters.

We measure the cumulative response to the field, not crater damage.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth: ~ 40 nm at 15 kV from Monte Carlo (CASINO), Drouin et al.,

Scanning 29, 92 (2007); measured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011).

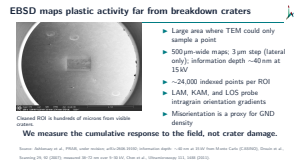
- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only); information depth ~ 40 nm at 15 kV
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for GND density

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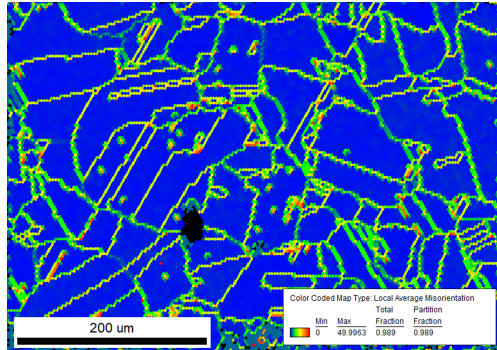
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└ EBSD maps plastic activity far from breakdown craters

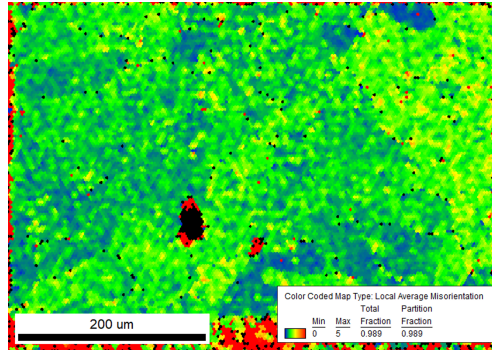
EBSD chosen for large-area where TEM is local. Shallow FIB clean removes oxide (under 10 nm by AFM), then map orientation gradients. The 3 micron step and kernel fix only lateral resolution; backscattered electrons come from top 40 nanometers of copper at 15 kV, so this is near-surface measurement. All ROIs deliberately far from craters, measuring cumulative material response to the field, not arc damage. LAM is primary metric; KAM and LOS check the result does not depend on one analysis choice.



How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate



0–5° scale: intragrain curvature

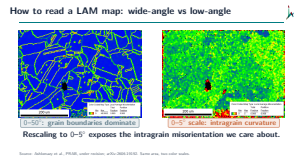
Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same area, two color scales.

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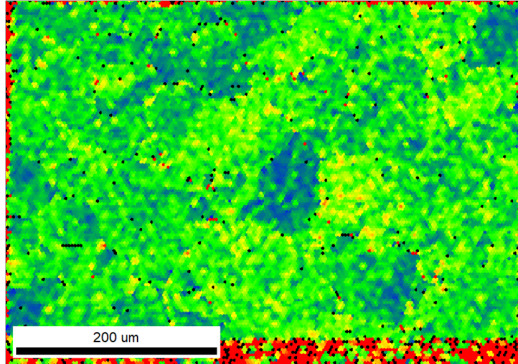
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How to read a LAM map: wide-angle vs low-angle

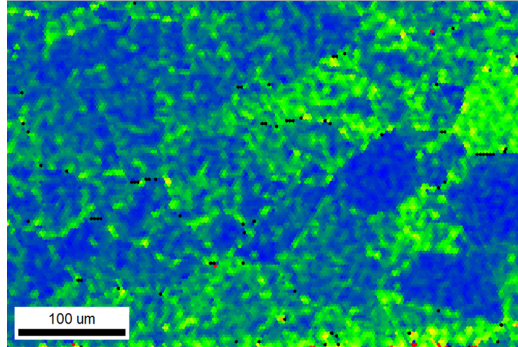


Teach the metric before comparison. At wide 0-50 degree scale only grain boundaries are visible. Rescaling to 0-5 degrees reveals intragrain lattice curvature—the low-angle misorientation proxying geometrically necessary dislocations. Every LAM map following uses this same 0-5 degree scale.

High-field copper contains more intragrain curvature



Field-exposed center



Unexposed reference

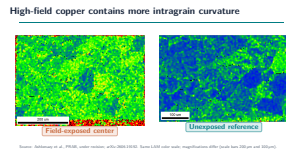
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same LAM color scale; magnifications differ (scale bars 200 μm and 100 μm).

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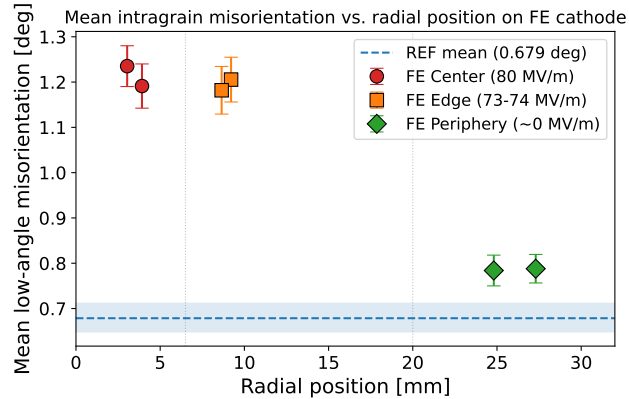
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└ High-field copper contains more intragrain curvature

Look at the spatial texture, not just average color. Field-exposed region carries far more high-misorientation structure inside the grains. Pair shown is representative; quantitative conclusion uses all nine ROIs.



The signal separates into three exposure tiers



- Center and edge: $\sim 1.2^\circ$
- External reference: $\sim 0.68^\circ$
- High field is $\sim 75\%$ higher
- Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

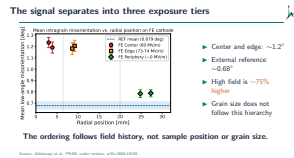
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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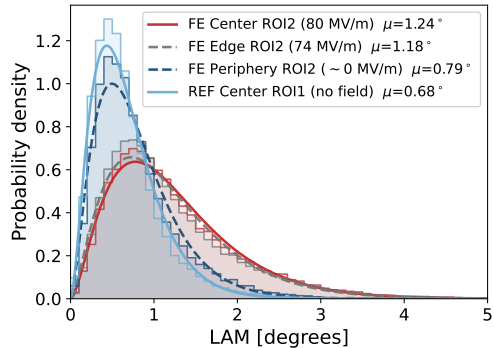
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The signal separates into three exposure tiers

Three tiers: high-field center and edge highest and similar; field-exposed periphery intermediate; unexposed references lowest. Periphery is powerful: it shared machining, heat treatment, mounting, vacuum, and handling with high-field regions but saw at most 2.5 MV/m—a low-field control, not zero-field. Intermediate value consistent with residual fringe-field exposure. The 75 percent figure characterizes this cathode and history, not a universal constant.



Field exposure builds a high-misorientation tail



LAM tail above 2°

~0.14 field-exposed
~0.016 reference
~8× increase

Gamma shape $k \approx 2.7$ fixed; the
scale roughly doubles.

The whole distribution stretches; a uniform offset cannot do that.

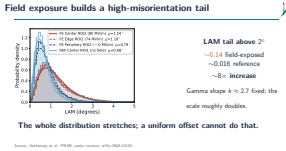
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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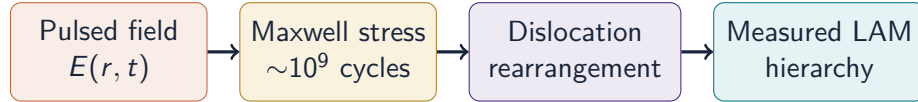
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Field exposure builds a high-misorientation tail

Low-angle distribution broadens and grows a long tail: fraction above two degrees rises roughly eightfold. Gamma shape parameter stays near 2.7 while scale roughly doubles—a fixed-shape stretch, signature of progressive deformation. LAM, KAM, and LOS agree. KS, chi-squared, and Anderson-Darling separate the distributions; every center or edge ROI differs from unexposed ones with KS distance above 0.25. Quote effect sizes and grain-corrected errors rather than extreme pixel-pair p-values.



The large-area test supports the model's prediction



- ▶ To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, as MDDF predicted.
- ▶ No conflict with the 2025 denuded zone: one mechanism, two starting populations (hard Cu depletes; heat-treated Cu builds curvature).
- ▶ The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

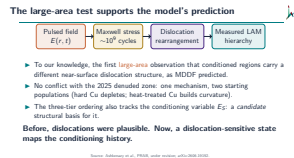
Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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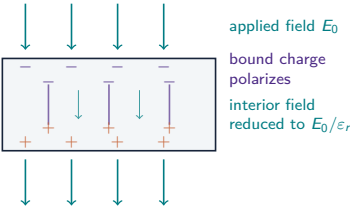


The measured hierarchy is the large-area structural signature the model predicted and STEM saw locally. Surface denuded zone and EBSD curvature are not in conflict: one mechanism acting on two very different starting populations. In dense as-machined copper, cyclic stress lets near-surface dislocations escape or annihilate—the depleted layer TEM saw. In heat-treated copper measured here, little to deplete, so same load generates and rearranges dislocations—the curvature EBSD records. Next slide gives length scale; depth connection still open. Only now, only in one line, connect to E_S : spatial ordering matches simulated E_S profile, so dislocation population is a candidate structural basis. Never say we measured E_S .

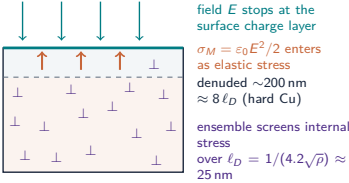
Elastic screening: dislocations respond like bound charge



Dielectric in an applied field



Copper under the Maxwell traction



applied field $\leftrightarrow$ Maxwell traction · bound charge $\leftrightarrow$ dislocations (dipole regime) · Debye length $\leftrightarrow \ell_D$ · screened interior field $\leftrightarrow$ screened internal stress of the depleted layer

2D single-slip theory at $\rho \sim 10^{14} \text{ m}^{-2}$; no theory yet treats cyclic loading with a free surface, so the $\sim 200 \text{ nm}$ thickness is observed, not predicted.

Charge screens the field at the surface. Dislocations screen the internal stress of a depleted layer, over $\sim 25 \text{ nm}$.

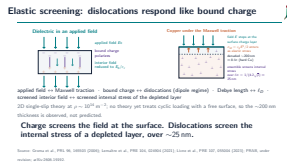
Source: Groma et al., PRL 96, 165503 (2006); Lemaître et al., PRE 104, 024904 (2021); Livne et al., PRE 107, 055004 (2023); PRAB, under revision; arXiv:2606.19192.

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Elastic screening: dislocations respond like bound charge

Picture dislocation population as bound charge of a dielectric: Maxwell traction is applied field, dislocations rearrange in response. In Groma's Debye-like theory, internal stress of any depleted or enriched layer is screened by surrounding ensemble over about 25 nanometers for as-machined copper. Contrast on the right: electric field itself screened by surface charge within a nanometer, never enters the metal; load enters as elastic stress, so electrostatic screening length says nothing about depth. STEM denuded zone in hard copper is about eight screening lengths thick; no theory yet treats cyclically loaded free surface, so thickness is observation, not prediction.



Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Call for cooperation: more cathodes, other materials, other facilities, other probes of the conditioned state (cf. M. Coman, 11:30) — let's compare notes.

A stronger test changes both the material and the loading history.

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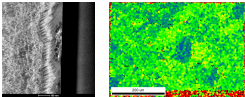
The honest limitations: one conditioned cathode and one external reference; field level and breakdown density covary; LAM is a relative proxy. Invite collaboration on replication and cross-facility comparisons. Depth matters: STEM depletion in hard copper, EBSD curvature in heat-treated copper, each in top few hundred nanometers at most. Depth profile on one material would show whether two ends of one redistribution. Next experiments add depth and change material and loading mode.

Next: go deeper, and change the material



Depth

- ▶ Matched high-field and reference FIB lamellae; TEM/STEM + EBSD through the top $\sim 1\mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?



Depth continues the Uppsala STEM line; material tests the mechanism itself.

Material & loading

- ▶ Stainless steel: lower stacking-fault energy, different dislocation mobility and slip
- ▶ Continuous DC instead of pulsed loading
- ▶ Possible metastable phase transformation



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Source: Interim work plan (2026); RFX project plan.

Two orthogonal directions. Depth is direct continuation of STEM work: pilot lamellae from mapped high-field ROI and matched reference with artifact controls learned in 2025. Material axis is RFX stainless-steel collaboration, changing stacking-fault energy, dislocation mobility, and loading mode from pulsed to continuous DC.

Stainless steel: a test chosen to be hard on the model



Copper
FCC, higher SFE
pulsed DC, $\sim 10^9$ cycles

AISI 304L SS
metastable austenite, lower SFE
continuous DC, possible $\gamma \rightarrow \alpha'$

Design: apex-to-periphery gradient; two exposed electrodes, two unexposed controls.

SS tracks field cross-material structural state

SS differs material-specific carriers

no resolved change informative only with enough maps

Status: 90 h and 73 h DC, peak 59–61 MV/m, zero breakdowns (stopped at emission switch-on); all 26 EBSD maps in hand; the exposed apex sits inside the unexposed range on every quantity tested, so no field indication yet.



Consorzio RFX — De Lorenzi, Pilan, Spada

Any outcome constrains the mechanism; the test is not pass-or-fail.

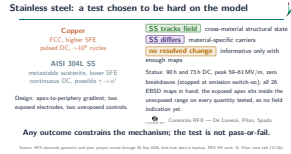
Source: RFX electrode geometry and plan; project record through 29 Sep 2026; first-look data in backup. RFX HV work: N. Pilan, next talk (12:30).

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└ Stainless steel: a test chosen to be hard on the model

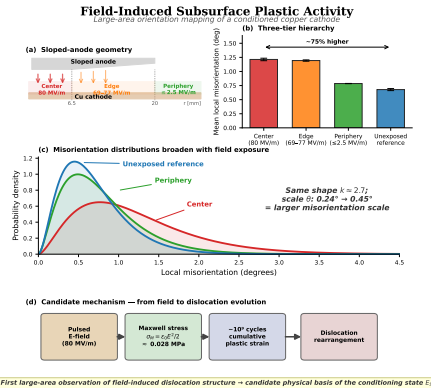
Stainless steel is hard: changes stacking-fault energy, dislocation mobility, loading mode, and adds possible austenite-to-martensite transformation. That makes it discriminating. Field-correlated response supports broad structural mechanism; qualitatively different response identifies material degrees of freedom storing state; null result is informative only once statistics bound sensitivity—current two to four maps per group cannot. With enough maps it may separate cyclic from continuous loading. Current status: two exposed electrodes ran 90 and 73 hours continuous DC at up to 61 MV/m, zero breakdowns. All 26 EBSD maps in hand; exposed apex sits inside unexposed range on every quantity tested, so no field indication yet. Runs stopped at emission switch-on. Hypothesis to test: as-manufactured 304L already in conditioned dislocation state, so run never engaged conditioning. Early apex-versus-side contrast reproduced on never-installed reference, so it is forming geography. No stainless-steel field effect claimed. First-look SEM and EBSD in backup.



Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** (~75% higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

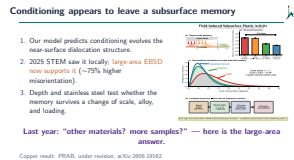
Copper result: PRAB, under revision; arXiv:2606.19192.

2026-10-02

Does conditioning leave a structural memory?

Conditioning appears to leave a subsurface memory

Close the loop from title and from last year. Model predicted structural memory; STEM saw it at one spot; large-area EBSD supports it across electrode with 75 percent effect and three-tier hierarchy. Stainless-steel and depth programs are next round of comparative forensics promised in 2025. Invite collaboration on more cathodes and cross-material tests. Cite manuscript as under revision at PRAB, arXiv:2606.19192.

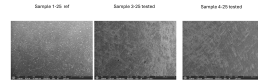


Backup: RFX first look (preliminary)

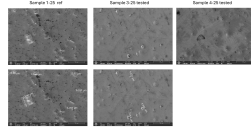
Preliminary — no field effect claimed

- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25: two maps, one an outlier on confidence index, position unrecorded
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography

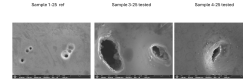
Source: RFX project record through 29 Sep 2026; SEM report, Inna Popov (20 May 2026).



Ref vs field-exposed overview



Sectioned electrode



2026-10-02

Does conditioning leave a structural memory?

└ Backup: RFX first look (preliminary)

Backup for questions on the stainless-steel first look. SEM survey plus EBSD quality: three samples index well; the field-exposed #4-25 has two maps, one of them an outlier on confidence index, and its position on the electrode was not recorded, so the cause is open. The July apex-versus-side contrast reproduced on a never-installed reference, so it reflects forming geography. No structural field effect is claimed.

Backup: RFX first look (preliminary)

Preliminary — no field effect claimed

- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean
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- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography

Source: RFX project record through 29 Sep 2026; SEM report, Inna Popov (20 May 2026).

Backup: RFX sample and data status (29 Sep 2026)



Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalogue EBSD maps (36 scans), 28 Sep.
- ▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
- ▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.
- ▶ Still open: thermal history of #3/#4 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM θ -survey as the first exposure marker.

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Does conditioning leave a structural memory?

└ Backup: RFX sample and data status (29 Sep 2026)

Use only if asked about readiness or missing inputs. Reflects the written project record through 29 September 2026.

Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

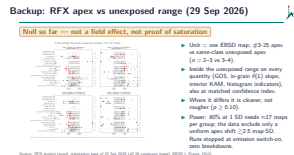
- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalogue EBSD maps (36 scans), 28 Sep.
- ▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
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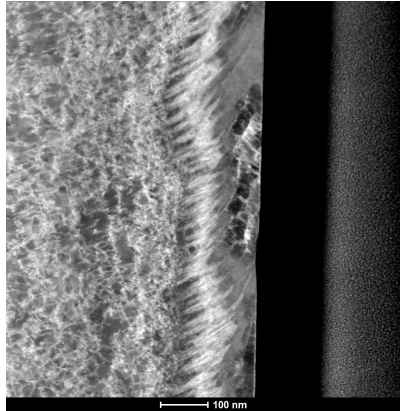
└ Backup: RFX apex vs unexposed range (29 Sep 2026)



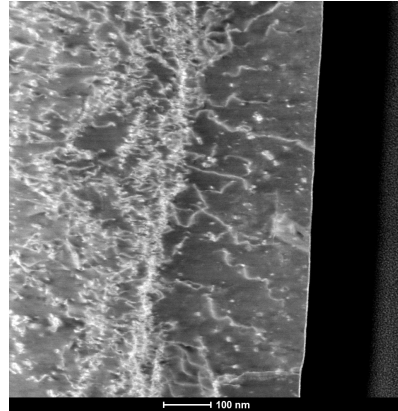
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Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls



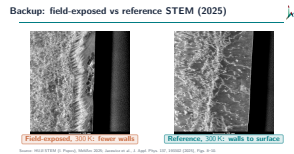
Reference, 300 K: walls to surface

Source: HUJI STEM (I. Popov), MeVArc 2025; Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.

2026-10-02

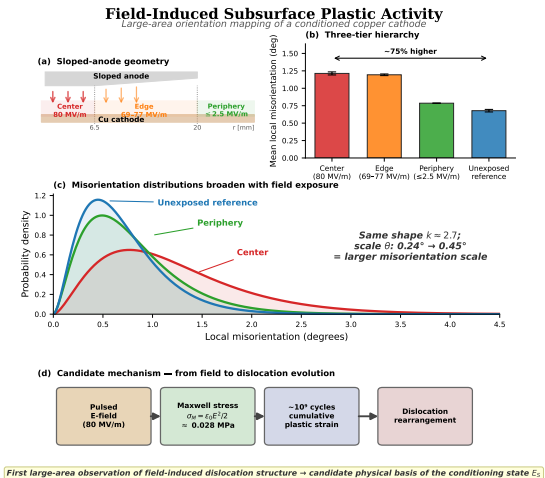
Does conditioning leave a structural memory?

└ Backup: field-exposed vs reference STEM (2025)



The field-exposed versus reference STEM pair from the 300 K cathode. In the published paper this pair shows a strong reduction of dislocation walls (Fig. 9); the zone the paper calls dislocation-denuded and clearly visible is the 30 K cathode (Fig. 10). No thickness is stated in the paper.

Backup: copper result in one figure



2026-10-02

Does conditioning leave a structural memory?

Backup: copper result in one figure

Compact answer to questions about geometry, distributions, or the proposed mechanism.

